

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `wave 2.3` residual harmonic temporal hybrid campaign into the official `TE Curve Verification Pipeline` offline verification package.

Decision

- `wave 2.3` is verified as an exploratory residual harmonic temporal baseline.
- No `wave 2.3` candidate is promoted over the current `wave 2.2` periodic sequence leaders.
- The accepted paper-derived forward baseline remains `paper_retuned_best_Fw`.
- The accepted paper-derived backward baseline remains `paper_retuned_best_Bw`.
- The strongest repository-owned static `TE Curve Verification Pipeline` baseline remains `tree`.
- The strongest repository-owned neural branch remains `wave 2.2`
`periodic_gru_sequence_Bw` for backward-only evaluation and `periodic_gru_sequence_global` for bidirectional neural evaluation.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28/validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28/per_condition_metrics.csv` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-28]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-28]/track2_multi_model_curve_comparison_report.md` .

Candidate Refresh

The refresh added 18 registry-backed wave 2.3 candidates:

Family	Surfaces
residual_harmonic_gru_sequence_sparse_rcim	global , Fw , Bw
residual_harmonic_gru_sequence_dense240	global , Fw , Bw
residual_harmonic_gru_sequence_dense360	global , Fw , Bw
residual_harmonic_lstm_sequence_sparse_rcim	global , Fw , Bw
residual_harmonic_lstm_sequence_dense240	global , Fw , Bw
residual_harmonic_lstm_sequence_dense360	global , Fw , Bw

The matrix now contains 111 candidates. The incremental operator run used the completed wave 2.2 refresh as its 93 -candidate baseline and evaluated only the 18 new wave 2.3 candidates.

Current Leaders

Scope	Current strongest candidate	MAE [deg]	RMSE [deg]	Mean [%]
forward overall	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372
backward overall	periodic_gru_sequence_Bw	0.002392	0.002639	5.466
paper-derived forward	paper_retuned_best_Fw	0.001839	0.002041	4.109
paper-derived backward	paper_retuned_best_Bw	0.003675	0.004284	7.572
repository static backward	tree_Bw	0.003258	0.003651	7.051
repository global static	tree_global	0.003144	0.003533	6.854
repository global neural	periodic_gru_sequence_global	0.002704	0.002949	6.139

Wave 2.3 Result

The strongest wave 2.3 candidates are

Scope	Strongest Wave 2.3 candidate	MAE [deg]	RMSE [deg]	Mean [%]
forward	residual_harmonic_gru_sequence_sparse_rcim_Fw	0.003194	0.003499	7.083
backward	residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.003440	0.003793	7.510
global	residual_harmonic_lstm_sequence_sparse_rcim_global	0.003368	0.003719	7.409

The sparse RCIM harmonic set is clearly the best wave 2.3 setting. The dense 240 and dense 360 residual harmonic variants are not competitive in the official matrix and appear to over-expand the harmonic base for this residual sequence branch.

Visual Evidence

The refreshed best-model collage report includes dedicated wave 2.3 sections for forward, backward, and global candidates. The refreshed multi-model overlay report includes dedicated wave 2.3 overlays and direct reference/tree versus wave 2.3 overlays.

Visual bundle paths

- doc/reports/analysis/track2/best_model_collage_report/[2026-05-28]/track2_best_model_collage_report.pdf ;
- doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-28]/track2_multi_model_curve_comparison_report.pdf .

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-05-28	wave 2.3 residual harmonic temporal hybrid refresh	18 global , Fw , and Bw residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with wave 2.3 sections	verified exploratory baseline; not promoted over wave 2.2 or accepted TE Curve Verification Pipeline baselines
2026-05-26	wave 2.2 harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch
2026-05-24	wave 2.1 temporal entry refresh	temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	visual reports refreshed	verified exploratory baseline
2026-05-21	periodic_mlp explicit harmonic registry refresh	global , Fw , Bw visual verification	source matrix unchanged; visual registry refresh included	collage and overlay PDFs refreshed	included as visual evidence, not promoted over tree

Closeout Decision

wave 2.3 does not change the accepted TE Curve Verification Pipeline baseline. Its practical value is diagnostic: it confirms that adding a residual recurrent branch over the structured harmonic base is viable only with the sparse RCIM harmonic set, while the dense harmonic banks are dimensionally expensive and less accurate.

The next modeling decision should keep wave 2.2 periodic sequence models as the strongest neural branch and treat wave 2.3 as a verified exploratory baseline rather than the next promotion target.

Curve-First Follow-Up

The next approved analysis direction is curve-first reranking before any new training campaign. The reason is practical compensation: future deployed models must follow complete TE curves for many consecutive motor revolutions, not only reduce pointwise scalar error on the dataset.

Follow-up rules

- keep this official report as the accepted `Wave 2.3` TE Curve Verification closeout;
- do not reopen `Wave 1` , `Wave 2.1` , `Wave 2.2` , or `Wave 2.3` ;
- create a separate `CVP 1.1 Curve-First Reranking` branch that evaluates accepted candidates on expanded full-curve metrics;
- preserve causal runtime inputs: current point-level operating state, optional short past-history windows, and derived causal features only;
- separate scalar registry winner, curve-first offline winner, and future deployment-ready candidate in later summaries.